Paper id: 252682

Roll No: | | | | | | | | | | | | | |

**BPHARM**
**(SEM IV) THEORY EXAMINATION 2024-25**
**MEDICINAL CHEMISTRY I – THEORY**

**TIME: 3 HRS** **M.MARKS: 75**

**Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.

## SECTION A

**1. Attempt *all* questions in brief.** **10 x 2 = 20**

| | |
|---|---|
| a. | Define bioisosterism and give one example. |
| b. | What are Phase I reactions in drug metabolism? |
| c. | Write the structure and use of phenylephrine. |
| d. | Give SAR of beta-blockers in brief. |
| e. | Differentiate between reversible and irreversible cholinesterase inhibitors. |
| f. | Write the synthesis and use of Neostigmine. |
| g. | What are barbiturates? Write any two examples. |
| h. | Write the mechanism of action of phenytoin. |
| i. | Mention the SAR of morphine analogues. |
| j. | Write the structure and therapeutic use of ibuprofen. |

## SECTION B

**2. Attempt any *two* parts of the following:** **2 x 10 = 20**

| | |
|---|---|
| a. | Describe in detail the physicochemical properties in relation to biological activity. |
| b. | Classify adrenergic agonists in detail . Explain the SAR of sympathomimetic agents with examples. |
| c. | Discuss the synthesis, mechanism, and SAR of cholinergic blocking agents. |

## SECTION C

**3. Attempt any *five* parts of the following:** **7 x 5 = 35**

| | |
|---|---|
| a. | Discuss the factors affecting drug metabolism including stereochemical aspects. |
| b. | Explain the biosynthesis and catabolism of catecholamines. |
| c. | Describe the SAR of benzodiazepines and write the structure of diazepam. |
| d. | Classify antipsychotic drugs. Explain SAR of phenothiazines. |
| e. | Write a short note on mechanism of action and synthesis of carbamazepine. |
| f. | Describe the classification and uses of general anesthetics. |
| g. | Discuss the structure, uses, and SAR of NSAIDs with special reference to ibuprofen and mefenamic acid. |

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